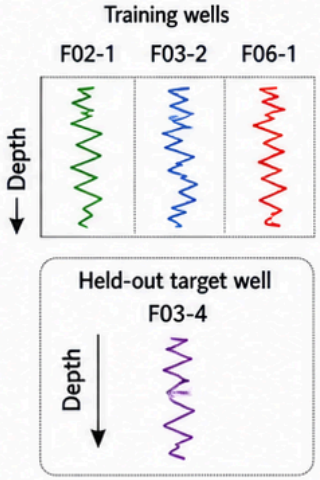
Experimental setup and evaluation workflow for CLP-CSGM

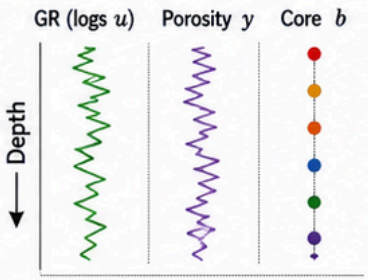
1. Data and experimental regimes

A Cross-well low-data benchmark



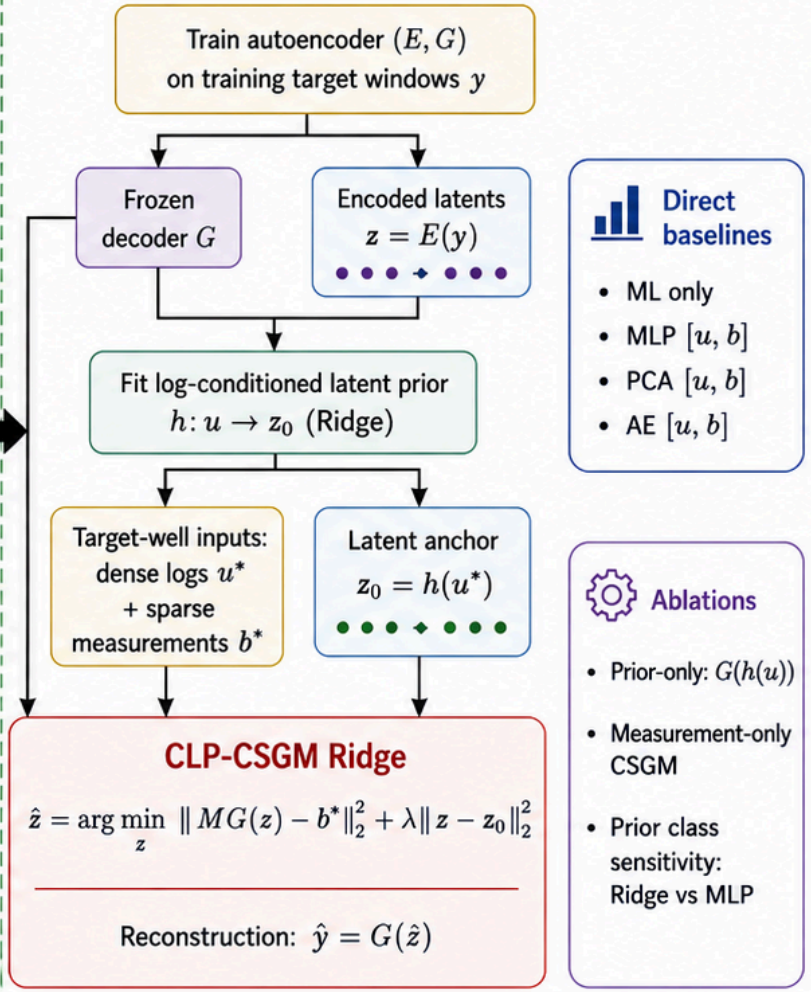
- Training wells: F02-1, F03-2, F06-1
- Held-out target well: F03-4
- Dense logs u
- Target property $y = V_c$
- Windowing + normalization
- Low-data settings via step $\{8, 16, 32\}$
- Sparse measurements generated by subsampling the target profile
- Measurement ratio ρ in $\{0.05, 0.10, 0.20\}$

B Real-well F03 benchmark



- Dense logs u (GR-only stress test)
- Target property: porosity
- Sparse core measurements b
- Measurement ratio ρ in {0.20, 0.30, 0.40, 0.50, 0.60}

2. Model construction and inference



Direct baselines

- ML only
- MLP $[u, b]$
- PCA $[u, b]$
- AE $[u, b]$

Ablations

- Prior-only: $G(h(u))$
- Measurement-only CSGM
- Prior class sensitivity: Ridge vs MLP

3. Experimental sweep and evaluation

Cross-well sweep

- step in $\{8, 16, 32\}$
- ρ in $\{0.05, 0.10, 0.20\}$
- seeds $\{7, 23, 41\}$

Real-well sweep

- ρ in $\{0.20, 0.30, 0.40, 0.50, 0.60\}$
- seeds $\{7, 23, 41\}$

Metrics: RMSE, MAE, relative L_2

Diagnostics: parity plots, profile overlays, runtime vs error

Aggregate tables and figures